

Nobel Prep

SAT Practice Exam

Mock Test 1

Math — Module 2

22 Questions · 35 Minutes · Calculator Permitted

Source	June 2023 Form B
Section	Math
Module	2 of 2
Time Allowed	35 minutes
Questions	22
Student Name	
Date	

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions. A reference sheet, calculator, and these directions are always available.

Multiple-choice questions — Choose the best answer. The correct answer is one of the choices given.

Student-produced response questions — Enter your answer in the box. Answers may be positive or negative numbers, decimals, or fractions. Do not enter symbols such as a percent sign, comma, or dollar sign.

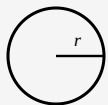
DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions.

NOTES

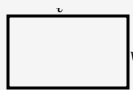
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

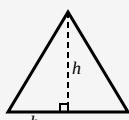
REFERENCE

$$A = \pi r^2$$

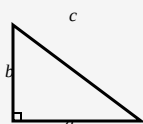
$$C = 2\pi r$$



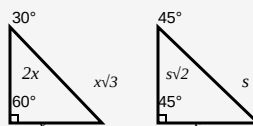
$$A = \ell w$$



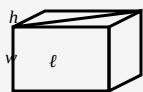
$$A = \frac{1}{2}bh$$



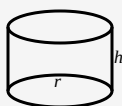
$$c^2 = a^2 + b^2$$



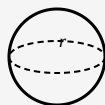
Special Right Triangles



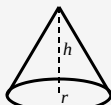
$$V = \ell wh$$



$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

DIRECTIONS

The questions in this section address a number of important math skills. Use of a calculator is permitted for all questions. Unless otherwise indicated: all variables and expressions represent real numbers; figures provided are not drawn to scale; all figures lie in a plane; the domain of a given function is the set of all real numbers for which the function has real values.

- 1 The speed of a white-throated needletail, a type of bird, in flight was measured to be 49 miles per hour. What was the white-throated needletail's measured speed, in kilometers per hour? (Use 1 mile = 1.6 kilometers.)

(A) 30.6
(B) 47.4
(C) 50.6
(D) 78.4

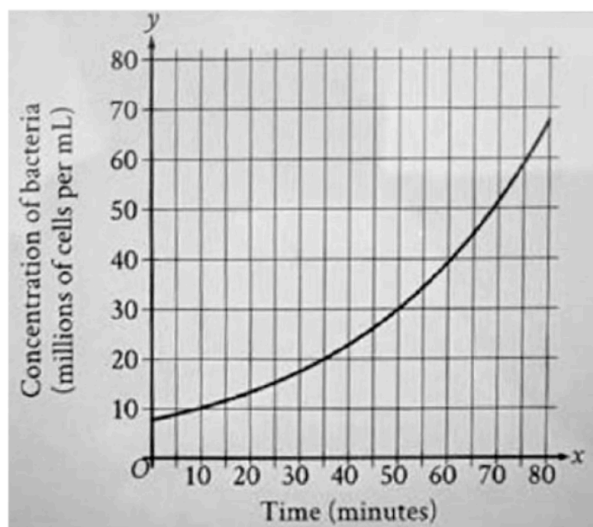
- 2 Which expression is equivalent to $(3x^3 - x^2 + 4)(5x^2 + 8x)$?

(A) $15x^5 + 19x^4 - 8x^3 - 20x + 32$
(B) $15x^5 + 19x^4 - 8x^3 - 20x^2 + 32$
(C) $15x^5 + 19x^4 - 8x^3 + 20x^2 + 32x$
(D) $15x^5 + 29x^4 - 8x^3 + 20x^2 + 32x$

- 3 If $5(x + 1) = 25$, what is the value of $x + 1$?

(A) 4
(B) 5
(C) 19
(D) 20

4



The graph shows the estimated concentration of a certain strain of bacteria, in millions of cells per mL of nutrient medium, over time x , in minutes since the initial measurement. According to the graph, which of the following is closest to the number of minutes it took for the estimated concentration of the bacteria to increase from 20 million cells per mL of nutrient medium to 30 million cells per mL of nutrient medium?

- (A) 15
- (B) 20
- (C) 30
- (D) 35

5

If $(x + 3)^2 = 30$, what is the value of $x^2 + 6x$?

- (A) 21
- (B) 30
- (C) 39
- (D) 60

6

$$\begin{aligned}y &= 5x \\ y &= 2x + 2\end{aligned}$$

How many solutions does the given system of equations have?

- (A) Exactly one
- (B) Exactly two
- (C) Infinitely many
- (D) Zero

7

$$f(x) = |71 - 2x|$$

The function f is defined by the given equation. For which of the following values of k does $f(k) = 3k$?

- (A) $\frac{71}{5}$
- (B) $\frac{71}{2}$
- (C) $\frac{213}{5}$
- (D) 71

8

$$g(x) = 2(16x - 17)$$

What is the y -coordinate of the y -intercept of the graph of $y = g(x) - 3$ in the xy -plane?

- (A) -37
- (B) -34
- (C) -20
- (D) -17

9

x	y
0	n
4	$n + 19$
8	$n + 38$

There is a linear relationship between x and y . The table shows three values of x and their corresponding values of y in terms of a constant n . What is the slope of the line that represents this relationship in the xy -plane?

10

$$18x^2 + 24x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

11

$$(x - 3) - 8(y + 9) = 129$$

$$(x - 3) + 8(y + 9) = 432$$

The solution to the given system of equations is (x, y) . What is the value of $8(x - 3)$?

12

The volume of a right rectangular prism with a square base is 2448 cubic centimeters. If the area of the square base is 144 square centimeters, what is the area, in square centimeters, of one of the four lateral faces of the prism?

- (A) 17
- (B) 204
- (C) 540
- (D) 816

13

$$y = 5(x - 2)^2$$
$$y = 10(x - 2)$$

A solution to the given system of equations is (x, y) . What is one possible value of $(x + y)$?

- Ⓐ -20
- Ⓑ 4
- Ⓒ 20
- Ⓓ 24

- 14 The graph of the quadratic function $y = f(x)$ in the xy -plane intersects the x -axis when $x = 39$ and when $x = p$, where p is a constant. The maximum value of $y = f(x)$ occurs at the point $(14, m)$, where m is a constant. What is the value of p ?

- 15 In the given equation, k is a constant. The solution to the given equation is $\frac{1}{174}$. What is the value of k ?

- 16 The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 6$ centimeters, and the height of the triangle is $x - 2$ centimeters. What is the value of x ?

- 17 The function g is a quadratic function. In the xy -plane, the graph of $y = g(x)$ has a vertex at $(-1, -4)$ and passes through the points $(-2, -43)$ and $(1, -160)$. What is the value of $g(0) - g(2)$?

- (A) -121
(B) 0
(C) 117
(D) 312

- 18 Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 62° . What is the measure of angle F ?

- (A) 28°
(B) 62°
(C) 90°
(D) 118°

- 19 A business consultant charges \$408 for the first hour and \$204 for each additional hour of work. Which of the following functions gives the charge C , in dollars, for h hours of work, where h is a positive integer?

(A) $C(h) = 204h + 204$
(B) $C(h) = 204h + 408$
(C) $C(h) = 408h + 204$
(D) $C(h) = 408h + 612$

- 20 An acceptable noise criterion rating for the background noise in a laundry room is 50. For a noise criterion rating of 50, the equation $y = 22(0.997)^{x-60} + 47$ gives the estimated sound pressure level, y , in decibels, as a function of the octave band center frequency, x , in hertz, where $x \geq 60$. Which of the following is the best interpretation of 47 in this context?

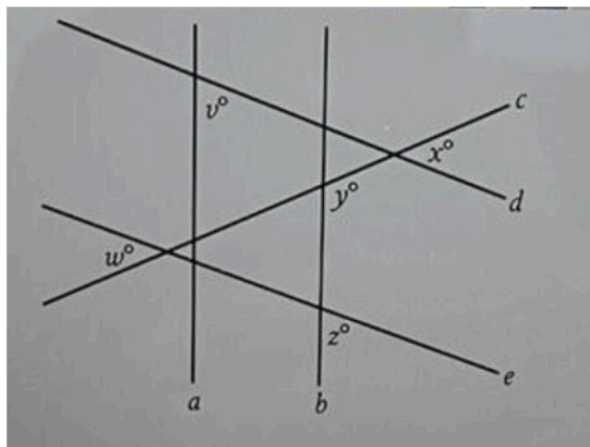
(A) 47 is 22 less than the estimated sound pressure level, in decibels, at an octave band center frequency of 60 hertz.
(B) 47 is 22 less than the estimated sound pressure level, in decibels, at an octave band center frequency of 0 hertz.
(C) 47 is the estimated sound pressure level, in decibels, at an octave band center frequency of 60 hertz.
(D) 47 is the estimated sound pressure level, in decibels, at an octave band center frequency of 0 hertz.

21

$$\begin{aligned} 3x + 5y &= 8 \\ 9x + 15y &= 24 \end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

22



In the figure, parallel lines a and b are intersected by lines c , d , and e . If $z = 49$, $y = 136$, and $v < z$, which statement about x and w must be true?

Answer Key — Math Module 2

Q1 D MCQ

Q4 B MCQ

Q7 A MCQ

Q10 8 SPR

Q13 D MCQ

Q16 6 SPR

Q19 A MCQ

Q22 $x + w = 180$ SPR

Q2 D MCQ

Q5 A MCQ

Q8 A MCQ

Q11 $1405/2$ SPR

Q14 -11 SPR

Q17 D MCQ

Q20 C MCQ

Q3 B MCQ

Q6 A MCQ

Q9 $19/4$ SPR

Q12 B MCQ

Q15 1 SPR

Q18 A MCQ

Q21 $(r, (8-3r)/5)$ SPR